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Neuropsychiatric Constructs as Bridges Between Psychopathology and Neuropathology: A Medical Perspective

Jesús Ramírez-Bermúdez[1] · Fernanda Pérez-Gay Juarez[2] · Atocha Aliseda[3]

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Abstract

This essay provides an analysis of the clinical problems that arise at the borderline between neurology and psychiatry. We postulate that psychopathological and neuropathological constructs have distinct referents and conceptual fields, but they are not mutually exclusive categories. After establishing criteria for defining neuropsychiatric constructs, we outline rules for identifying cases where a significant relationship exists between neuropathological and psychopathological patterns. We propose three approaches to establishing this relationship: a clinical epidemiology approach, a clinical neuroscience approach offering a mechanistic model, and a longitudinal study of individual cases incorporating idiographic perspectives. We discuss the logic of brain-behavior relationships and emphasize the need for closer academic feedback between neuroscientific research and clinical practice to better understand the causality of psychopathological phenomena in the context of neuropsychiatry. While precise demarcation may be challenging in some cases, we provide two types of examples: first, straightforward cases where neuropathology explains psychopathological patterns; second, complex cases that require studying the causal interactions between neuropathological factors, psychological factors, and sociocultural contexts, leading to intricate clinical patterns that demand cognitive neuropsychiatry resources, and a multidisciplinary approach to clinical care.

Keywords Neuropsychiatry · Epistemology · Psychopathology · Neuropathology · Philosophy of psychiatry

✉ Jesús Ramírez-Bermúdez
jesusramirezb@yahoo.com.mx

1 Neuropsychiatry Unit, National Institute of Neurology and Neurosurgery (Mexico), Insurgentes Sur 3877, Ciudad de Mexico, Mexico

2 Philosophy Department, McGill University, Montreal, Canada

3 Instituto de Investigaciones Filosóficas, Universidad Nacional Autónoma, Ciudad de Mexico, Mexico

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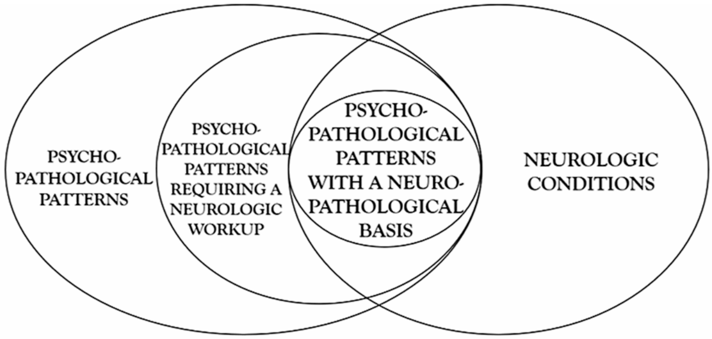
1 Introduction: The Borderland Between Neurology and Psychiatry

Neurology and psychiatry are institutionalized medical disciplines that arise as the result of an overwhelming demand of clinical care in most contemporary societies around the globe. As clinical research develops, the boundaries between neurological problems and psychiatric problems are unstable and unclear at the theoretical and practical levels. This article provides a conceptual approach to the scientific and clinical problems that arise in the borderland of neurology and psychiatry. We conceptualize neuropsychiatry as an interdisciplinary field of medicine that addresses the overlap between neurology and psychiatry. According to Alwyn Lishman, this field has two complementary aspects: academic neuropsychiatry and clinical neuropsychiatry. Academic neuropsychiatry is a theoretical field concerned with the scientific explanation of the full scope of psychopathological phenomena, encompassing the neurobiological study of primary psychiatric disorders and the psychiatric phenomena observed in neurological disease, and integrating advances from cognitive neurology, and biological psychiatry (Cummings and Hegarty 1994). On the other hand, clinical neuropsychiatry is a practical discipline —a form of medical practice—dedicated to the study and clinical care of patients with psychiatric symptoms caused by brain pathology that is demonstrable at the individual level (Lishman 1992). Figure 1 shows both concepts, which are complementary because clinical neuropsychiatry refers to medical practice and academic neuropsychiatry provides the scientific framework for explanation, prediction and intervention.

In this essay, we argue that psychopathological and neuropathological constructs have different referents and conceptual fields: in one case, we deal with clinical phenomena observed at the level of behavior and subjective experience; in the other case we deal with neurobiological factors through the resources of neuroscientific technology. But there is a causal interplay between the variables included in the two levels of analysis. We argue that the categories of neurology and psychiatry are not mutually exclusive, because an etiological relationship between the psychopathological and the neuropathological variables can be established for some cases. Not all psychiatric or neurological cases fall under the neuropsychiatric category; only the subset of them in which a neuropathological pattern is significantly related to a psychopathological pattern. While precise demarcation may be challenging in some cases, we offer a set of criteria to enhance the clinical identification of cases deemed neuropsychiatric, enabling multidisciplinary management.

As some philosophical debates arise from a terminological problem concerning the use of terms like “disease”, “illness”, “disorder”, and “pathology”, Sect. 2 provides a medical conceptualization of “the pathological.” This will be the starting point to provide characterizations of “the neuropathological” and “the psychopathological”, which are developed in Sect. 3 and Sect. 4. In Sect. 5, we define neuropsychiatric constructs according to three conditions: the recognition of a psychopathological pattern, an identified neuropathological pattern, and a significant relationship between the psychopathological and neuropathological patterns that can be established through scientific explanations. We analyze the types of explanations that are useful and valid in this context: a clinical epidemiology approach, a clinical neuroscience approach that provides a mechanistic model and connects with neurocognitive models of psy-



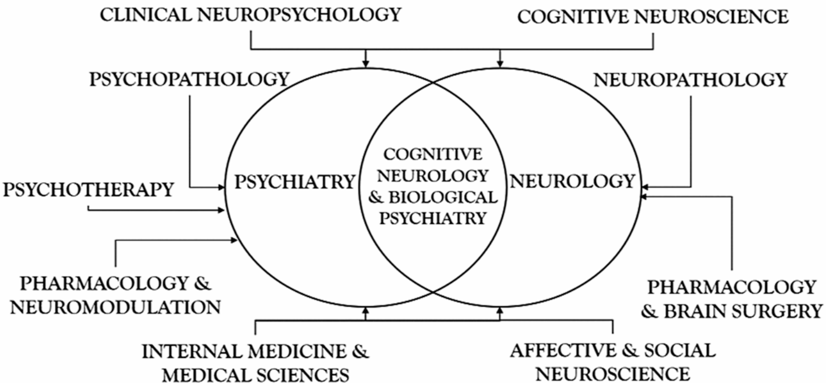


Fig. 1 The two sides of neuropsychiatry. Above, clinical neuropsychiatry is a practical discipline dedicated to the study and clinical care of patients with psychopathology caused by brain pathology which is demonstrable at the individual level. Below, academic neuropsychiatry is a theoretical activity concerned with the scientific explanation of the full scope of psychopathological phenomena, encompassing the neurobiological study of primary psychiatric disorders and the psychiatric phenomena of neurological disease

chopathology, and a longitudinal study of individual cases, allowing for causal intervention, functional analysis, and idiographic perspectives.

2 A Conceptual Approach to the Pathological

2.1 The Concept of Illness

To understand the pathological, we will use two complementary concepts: illness and disease. The pathological is commonly associated with the objective disturbance of bodily function and structure, but the dimension of illness is a relevant aspect of the pathological: one where medical practice and popular culture converge. By using the term illness, we refer, firstly, to the subjective state of suffering or distress. In Spanish language, we use the term *padecer*, which derives from *pathos*, just as the term "pathology" does. From an etymological point of view, *pathos* (from ancient Greek, πάθος) refers to ailment, suffering; in a broader sense, the term was used in Greek antiquity to refer to being affected by feelings, according to Aristotle's *Rhetoric*. There is no perfect translation for the Spanish term *padecer*, so we are using "illness" in the sense provided by Arthur Kleinman: the human experience of symptoms and suffering, including how individuals enact this experience: monitoring bodily processes, evaluating their meaning, and finding ways to cope (Kleinman 1988). In the words of a general practitioner, "Illness is a feeling, an experience of unhealth which is entirely personal. Often it accompanies disease, but the disease may be undeclared, as in the early stages of cancer or tuberculosis or diabetes. Sometimes illness exists where no disease can be found." (Marinker 1975). This core feature of medicine could be more relevant in psychiatry: many psychiatric problems can be conceived as "illness" without always being able to diagnose a disease. "The old aphorism that there are no diseases, only patients, precisely alludes to this confrontation between disease and illness," says Lifshitz (Lifshitz 2008). According to this author, nosological diagnosis is the act of giving the illness the name of a disease. It is necessary, therefore, to analyze the notion of disease.

2.2 The Concept of Disease

As there is no singular and universal definition of the term "disease," we summarize here a preliminary set of useful criteria for what we will frame as the anatomopathological and pathophysiological concept of disease. We should clarify that we are not trying to preclude other technical or folk uses of the word "disease". These preliminary criteria are intended only as a resource to analyze whether a process or condition can be understood as a disease, from the perspectives of clinical pathology and/or clinical physiology, even if exceptions can be found for each criterion.

(1) A disease is a condition that alters health, as defined by any of these features:

   a. A significant disruption of the physiological processes that are required to guarantee the organism's homeostatic balance, as may be verified by means of the direct clinical examination, laboratory tests, clinical physiology tests, imaging tests, or other technological resources with diagnostic validity according to current medical science.
   b. A cellular damage in one or more tissues of the body, as may be verified by means of the direct clinical examination, the use of imaging technologies, or through a direct pathological analysis of the tissues.

(2) The disturbance in health as defined in (1) has relevant consequences for the organism's life, for instance:

   a. A significant increase in the risk of premature death.
   b. Functional disability.

c. A change of the organism's interaction pattern in a way that jeopardizes the organism's integrity and/or those around them.
d. A significant loss of quality of life and psychological well-being.

2.3 The Progressive Conceptualization of Pathological Problems According to Medical Research

To understand pathology, we need to consider the progressive levels of scientific conceptualization and diagnosis. Table 1 provides an adaptation of Ronald S. Pies' proposal, (Pies 2009) in which we have incorporated some terms: symptom, sign, syndrome, disorder and disease. Diagnosis dynamically evolves through the clinical method; it involves a process of abductive reasoning combined with deductive reasoning when discarding diagnostic hypotheses. Following C.S. Pierce, the abductive process can be characterized as involving stages of discovery, hypothesis formation and testing, belief updating, concluding with an explanatory formulation of an unusual fact, based on the clinical variables in play and the background scientific theory (Aliseda 2006). Therefore, the diagnostic process can be conceptualized as a

Table 1 The progressive scientific conceptualization of the pathological evolves from the notion of "symptom", which would be the basic unit of subjective discomfort arising in the context of the pathological, to the concept of a well-defined disease from an anatomopathological and/or pathophysiological standpoint. Adapted with permission from the model by Pies (2009)

STAGE 5	FULLY REALIZED DISEASE ENTITY	ANATOMO-PATHOLOGIC & PATHO-PHYSIOLOGIC CONCEPT OF DISEASE	A clinical pattern with a fully known etiology and a specific molecular characterization, including a full phenomenological, molecular and epidemiologic description for all subtypes	INCREASING LEVELS OF PREDICTIVE VALIDITY. NATURAL KINDS? ↑
STAGE 4	SPECIFIC DISEASE		A clinical pattern with known etiology, specific biomarkers, well defined phenomenology, clinical course, subtypes, comorbidity, epidemiological features, treatment response	
STAGE 3	PROTO-DISEASE	PRIMARY DISORDER (IDIOPATHIC	A clinical pattern with well-defined phenomenology, clinical course, subtypes, comorbidity, epidemiological features, treatment response, preliminary pathophysiology	
STAGE 2	GENERAL SYNDROMAL DESCRIPTION	SYNDROME	A well-recognized clinical pattern of signs and symptoms which may be due to different diseases or different disorders	
STAGE 1	INITIAL INDICATORS OF SUFFERING, DYSFUNCTION AND/OR DISABILITY	SIGNS	Bodily/behavioral changes caused by pathologies & observed by clinical exams	
		SYMPTOMS	Subjective reports by the patient related to significant distress and/or impairment of the regular interactions, according to the premorbid standards of the patient	INCREASING LEVELS OF UNCERTAINTY. PRACTICAL KINDS?

nomothetic resource that bridges the individual case with general nosological categories. This conceptualization highlights the hypothetical nature of medical constructs. These hypotheses enable scientific research and science-based decision making (See Table 2).

In Table 1, the lower levels of diagnostic certainty align with **stage 1**, characterized by the presence of symptoms. These are subjective reports which serve as initial indicators of suffering, dysfunction, and disability. The recognition of signs, bodily or behavioral changes observed through clinical examination, enhances the objectivity of diagnostic reasoning. In **stage 2**, clinicians identify clusters of signs and symptoms, leading to the recognition of clinical patterns through a matching process. Key findings and the related clusters are used to compare these features with regards to clinical prototypes (Huda 2021). The syndromic diagnosis involves an initial synthesis where clinicians attempt to make sense of individual signs and symptoms. By definition, a syndrome is a cluster of signs and/or symptoms that may be caused by several alternative diseases. Further investigations are then necessary to provide a differential diagnosis, leading to more stable nosological diagnoses and progressing to **stage 4** or **stage 5**, which correspond to anatomopathological and/or pathophysiological diseases. These stages are defined by objective and specific forms of tissular damage or physiological abnormalities, which are demonstrable at the individual level, enabling higher levels of predictive validity. However, specific causes and biomarkers are not identified in many cases, leading to the diagnosis of a "primary disorder," referred to here as **stage 3**: proto-disease (Pies 2009). This term is not intended to state that there is no real disease inside the categories of this stage, but rather, to acknowledge that our scientific understanding of these categories is preliminary. In the fields of neurology and general medicine, **stage 3** may encompass numerous disorders such as migraine, trigeminal neuralgia, fibromyalgia, long-COVID, that lack a physiological or anatomopathological definition. In psychiatry, **stage 3** includes the cases receiving diagnoses of schizophrenia, bipolar disorder, and all of the so-called primary psychiatric disorders. A further differentiation of **stage 3** is required, as it includes disorders with a psychological origin, like posttraumatic stress disorder, and disorders in which the origin is idiopathic as the etiology is not fully understood (schizophrenia or autism).

The concept of disease corresponds to a general nosological category with different levels of validity in terms of etiology, diagnosis, prognosis, and treatment. From an etiological point of view, validity pertains to the assertion that the disease manifests as a component of a well-established system of causal relationships, elucidated through scientific analysis. The identification of causal factors is regarded as an explicit criterion to define a disease. In the context of medical science, a nomothetical definition of cause has been provided by clinical epidemiology: a factor is a cause of an event if its operation increases the frequency of the event (Elwood 2009). As pointed out by Mark Elwood in his *Critical Appraisal of Epidemiologic Studies and Clinical Trials*, there are at least four types of causes in the context of clinical epidemiology: (a) necessary causes, when the outcome occurs only if the causal factor has operated, (b) sufficient causes, when the operation of the causal factor always results in the outcome, (c) necessary and sufficient causes, when the cause and the outcome have a fixed relationship, and neither occurs without the other, and (d) contributing

Table 2 A systematic description of the structural changes, the physiological and the psychological aspects of two neuropsychiatric cases, providing the basis for a case formulation including a structural and functional analysis of the psychopathological patterns

Level of description	Case B	Case D
Structural lesions	Fronto-parietal white matter damage observed as punctate gliosis in the semioval center, bilaterally. Sequelae of microhemorrhages in the right putamen and the left globus pallidus	Hyperintense abnormalities in the medial aspect of the temporal lobe, bilaterally, probably related to acute inflammation
Physiologic abnormalities	Oxygen saturation levels of 60%, abnormalities in fronto-parietal executive network, neuroinflammation	Generalized slowing in the electroencephalographic study in the acute phase of the autoimmune disease
Behavior	Psychomotor slowing, suicidal behavior, isolation, insomnia, rejection of food	Catatonic behavior in the acute phase. Suicidal behavior in the long term
Subjective experience	Anhedonia, anergia, feelings of guilt, worthlessness, anguish, hopelessness	Auditory hallucinations
Neuropsychological Functions	Severe impairments in processing speed, working memory and in the ability to retrieve information in tasks involving semantic memory, episodic memory, and visual memory	Generalized and severe cognitive dysfunction in the acute phase
Psychological pattern	Anxious depression and moderate to severe cognitive impairment	Psychosis progressing to delirium and catatonia in the acute phase of encephalitis. Affective dysregulation and suicidal behavior in the long term
Biological causal factors	COVID-19 infection with involvement of the respiratory system and the central nervous system	Anti-NMDAR encephalitis
Psychosocial causal factors	Traumatic experience during the acute infection, death of two close family members in the same timeframe	Sexual abuse in childhood

causes, when the operation of the causal factor increases the frequency of the outcome, but the outcome does not always result, and the outcome can occur without the operation of the causal factor (Elwood 2009). Most frequently, human disease is the result of multicausal interactions (Elwood 2009).

Validity -in terms of diagnosis- means that the disease can be identified with accuracy and precision through a process of reasoning with the resource of clinical and paraclinical tests: it can be differentiated from other pathological and non-pathological conditions. In terms of prognosis, validity corresponds to the predictive capacity that clinicians acquire when establishing a diagnosis. Recognizing a disease implies the ability to make increasingly accurate and precise predictions about desirable and undesirable outcomes. Validity, in therapeutical terms, means that the diagnosis allows us to make decisions about a treatment (whether preventive, curative, symptomatic, or palliative). Once the concept of the pathological has been discussed in a general manner, it is necessary to move towards a more specific analysis of neuropathological and psychopathological constructs, which involve conceptual and practical challenges.

3 Neuropathological Constructs

It is generally accepted that the anatomical studies of Vesalius and the development of optical technologies giving rise to microscopy and neuro-histology were critical facts leading to the European revolution of anatomopathological medicine. The cellular pathology developed by Rudolf Virchow, Darwin's evolutionary theory, the physiological conceptualization provided by Claude Bernard, and the neuronal theory by Santiago Ramón y Cajal, provided a new scientific framework for clinical medicine. Along the XIX century and the early XX century, the anatomical pathology approach was successful to explain paradigmatic cases of behavioral or psychiatric disturbances. For example, language dysfunction related to lesions in the left hemisphere, as described by George Dax, Paul Broca, and Karl Wernicke, or presenile cases of dementia, by Alois Alzheimer (Ramirez-Bermudez 2012). Most of the mental disturbances that remained without a neurological explanation were discarded as part of the neuropathological field and remained within the psychiatric canon. In this essay, neuropathological constructs are defined as clinical patterns which are causally related to structural lesions and pathophysiological abnormalities of the nervous system. A brief explanation of this criteria should be provided.

3.1 Structural Lesions of the Nervous System

Classically, structural lesions should be observed by means of cellular pathology in the direct assessment of the tissue, by autopsy or biopsy. After the revolution provided by neuroimaging techniques, the use of computed tomography, magnetic resonance imaging and molecular imaging techniques became the practical standard for the recognition of structural lesions (although the direct pathological examination of tissues is still of critical value in many fields, as would be the case of neurooncology). Some neurological conditions -for example, Parkinson's disease- are generally diagnosed on a clinical basis, without the resource of biomarkers. However, they are not considered "functional" disorders because brain pathology studies in autopsy specimens demonstrate the relationship between the clinical pattern and specific structural abnormalities. The clinical pattern is regarded as a proxy for the "hidden" lesion, with a reasonable degree of validity. This view has the advantage of being open to etiological discovery and therapeutics. In a way, this criterion is consistent with the perspective that has been depicted as "the narrow view" of brain disorders (Jefferson 2022).

3.2 Pathophysiological Abnormalities

Following Claude Bernard, the physiological study of the milieu intérieur -or internal environment- became the biological framework to study the health and disease dynamic process. According to Bernard, the stability and constancy of the internal environment (homeostasis) is the condition for life, through the physiological and behavioral work of the organism to ensure the maintenance of the physical, chemical, and biological conditions within its internal environment, essential for cell survival. This presupposes that the systemic and highly coordinated activity of the organism compensates for the continuous variations in the external environment to ensure the dynamic balance of the internal milieu, by means of a brain-centered, predictive mode of physiological regulation (to achieve stability through change) which has been conceptualized as allostasis (Schulkin and Sterling 2019). The demonstration of a physiological abnormality or the determination of a specific biochemical or cytological biomarker is of value to establish causal inferences with regards to clinical patterns. Also, clinical neurophysiology involves the measurement of the electrical activity of the nervous system. This enables the recognition of abnormal neurophysiological patterns in epilepsy or encephalopathy. The biological marker is used to track a psychophysiological process connecting the etiological factors and the clinical pattern. Of note, this functional criterion for brain disease is compatible with the inclusive account of brain dysfunction provided by Anneli Jefferson as a criterion to support the notion of mental disorders as brain disorders (Jefferson 2022). From our perspective, however, to support this inclusive functional account with medical rigor the brain dysfunction should be demonstrable both at the group level and in the specific case that undergoes a diagnostic process.

4 Psychopathological Constructs

Once we defined neuropathological categories, let us consider the characteristics of a psychopathological pattern. Clinically, these are not defined by the presence or absence of structural lesions or physiologic abnormalities of the nervous system: psychopathological constructs are defined by recognizable clinical patterns at the level of the psychological, with features of clinically relevant subjective distress (suffering) and/or functional impairment. The concept of harmful dysfunction may be relevant to discriminate between psychopathological patterns and "problems in living" (Wakefield 2017).

As German Berrios wrote, the "psychological" is difficult to define due to its metaphorical origin (*Ψυχή*, *psyche*) and ontological instability (Berrios 2018). The word *psyche* was used in Classical Greece to mean "rush or air, blow, breath, and later -by dint of metaphor- was used to name the soul, conscious self, the source of life." (Berrios 2018) A similar problem arises with the definition of the "mental". The English word "mind" is related to Old English *Mynd*, Old High German *gimunt*, Gothic *gamund*, and to the latin term *mens*, which was the name a goddess (*Mens Bona*), the personification of "right thinking" (Adkins and Adkins 1998). Highlighting the metaphorical and mythological origins of the terms "psyche" and "mind" helps us to understand the relationship of these terms with cultural traditions and folk psychology. A philosophical perspective about the specific properties or marks of the mental is of great value, including the analysis of intentionality, consciousness, free will, teleology and normativity (Pernu 2017). Following Tim Crane, we regard intentionality as a technical term in philosophy that refers to the representational nature of mental states. In this context, intentionality means that mental states are about something—they refer to objects whether these objects are real, abstract, fictional or imaginary. Philosophers describe intentionality as the "aboutness" or "offness" of mental states, their directedness towards objects. To say that mental states exhibit intentionality is to assert that they are constituted by mental representations (Crane 1998, 2015) The problem of mental representation is of great importance to cognitive neuroscience, and scientific models of cognition must address at least three questions: To what degree can we regard cognitive processes as being representational, informational, and computational? In the context of medical, neuroscientific, and psychological practice and research, the word "psychological" does not entail a metaphysical dualistic view, in the sense of a soul-body substance-dualism. In this article, we will use the concept of "the psychological" to integrate three dimensions of analysis:

4.1 Behavior

Following a classic definition, we can start by saying that *behavior is what the organism is doing* (Skinner 1938). B.F. Skinner states that behavior "is only part of the total activity of an organism, and some formal delimitation is called for." Although the organism has multiple parts with multiple "actions" at the physiological level, it is better to restrict the concept of behavior interactions of the organism as a whole, as an integrated agent, and not to the specific physiological contribution of each part of the organism: "behavior is that part of the functioning of an organism which is engaged in acting upon or having commerce with the outside world" (Skinner 1938). Behavior may be regarded as the actions of the individual organism -the psychological agent- during the interplay with the environment, as this may be verified by an external observer (Skinner 1938). From an epistemological perspective, the clinical analysis of behavior falls under the terms of a third person perspective (Díaz 2016). In the context of psychopathology, behavioral signs are captured by external observers (for instance, in the case of catatonic signs or aggressive behavior) and thus they have a higher degree of objectivity as compared to symptoms.

4.2 Subjective Experience

While the examination of overt behavior from a third-person perspective is crucial for the logic of scientific objectivity, clinical practice must also take into account first-person and second-person accounts for pragmatic reasons (Díaz 2016). The scientific problem of consciousness lies at the core of clinical practice. The problem of subjective experience is the key aspect of consciousness (Seth and Bayne 2022). From a behavioral perspective, subjective experience can be framed as the result of private stimuli and covert behavioral responses, leading to private events (Tourinho 2006). Subjective phenomena remain inaccessible for public examination unless individuals communicate them through verbal language or other symbolic systems. The main method to access the patient's experience is by means of semi-structured or in depth-open interviews which reveal the subjective qualities of psychopathologic phenomena (Stanghellini and Broome 2014). The level of phenomenology brings a qualitative, subjective dimension to psychopathology, constituted by experiences that can be conceptualized as symptoms due to their distressful, unsettling, and atypical qualities, and because patients feel that these experiences are beyond their control, as disturbances of their regular mental experience and operation (e.g., obsessions, hallucinations, depersonalization). Communicating these experiences is consistent with the hypothesis according to which the functional role of subjective experience is related to intersubjective communication allowing for social organization and interpersonal care (Frith and Frith 2024).

4.3 Neuropsychological Processes

The concept of mental or neuropsychological processes is used here to categorize the operation of cognitive-affective functions of the nervous system which mediate the organism's sensory-motor interaction with the environment: the intermediary processing that is necessary to provide structure and content to subjective experience, and to organize the overt behavior. This processing includes both conscious and unconscious processes (Frith and Frith 2024). At the practical level, a "mental status exam" is undertaken in the clinical setting to assess the neuropsychology of alertness, attention, language (and the paralinguistic aspects of communication), memory, constructional abilities and other aspects of spatial cognition, numerical cognition, reasoning abilities, categorization and abstraction, sequencing and planning, decision-making processes, praxis, and the evaluative processes which have been traditionally conceptualized as "emotional or affective". The categories of mental function represent a problematic confluence of the philosophical, medical, psychological, and neuroscientific traditions. Thus, there are controversies regarding the scientific validity and the boundaries of these functions (Pessoa et al. 2022). It is expected that the continuous dialogue between basic, clinic and theoretical research will provide a scientific reconceptualization of cognitive-affective functions, to improve the mapping of mental function with regards to neurophysiological and neuroanatomical data, with the resources of evolutionary neurobiology (Cisek 2019).

4.4 Psychopathological Patterns

So far, we described three dimensions of analysis of the psychological: overt behavior, conscious or subjective experience, and neuropsychological functioning. These dimensions are integrated and thus we can speak about the global psychological activity of human agents in their ecological settings, allowing for meaningful and purposeful interactions with physical, chemical, biological, interpersonal, sociocultural, and technological environments. The patterns of interaction are the patterns of behavior, but the overt behavior cannot be understood and explained without considering the patterns of subjective experience and the patterns of neuropsychological functioning. Psychopathological patterns are constituted by clusters of behavioral signs and/or mental (subjective) symptoms that are often linked to neuropsychological dysfunctions, leading to significant issues in global psychological functioning. The relationship between neuropsychological dysfunctions and the emergence of psychiatric symptoms is a subject of both theoretical and empirical research, as postulated within the cognitive neuropsychiatry framework (Halligan and David 2001). Psychiatric symptoms such as hallucinations, delusions, or conceptual disorganization do not seem to result directly from a focal neuropsychological deficit. Instead, they could be viewed as indirect effects of neuropsychological abnormalities on distributed cognitive systems, leading to aberrant or distorted information processing due to a lack of inhibitory or modulatory control.

The process of clinical attention seeking usually starts with subjective complaints which can be regarded as mental symptoms, which refer to atypical, unsettling, distressful and problematic subjective events that have been incorporated by the tradition of psychopathology (for instance, hallucinations or feelings of depersonalization and derealization). According of Marková and Berrios, mental symptoms are “personal constructs” or personal interpretations of subjective experiences, described by the patient according to his/her catalogue of verbal categories, and influenced by personality factors, education, and sociocultural factors as well. Some clinical problems like psychomotor agitation, aggressive behavior, or catatonic signs are not subjective reports but instead they are overt behavioral events. To be conceptualized as signs and symptoms, these problematic behaviors and experiences need to be communicated by the patient and then judged as dysfunctional, harmful or pathological by the clinician. The historical, social and cultural factors in which the clinical relationship is immerse are variables influencing this diagnostic decision (Marková and Berrios 2009). However, there are also intrinsic variables which influence the diagnostic decision making, as the degree of suffering, how unusual the experience is, and the subjective feeling of impaired agency over the clinical phenomena. From our perspective, to be considered a clinical pattern, the cluster of behavioral signs and/or mental symptoms should align with the following profile:

A. A severe and/or prolonged state of suffering experienced by the patient as something outside of their voluntary control, and which does not improve with the conventional remedies provided by the popular culture surrounding the patient.
B. A significant impairment of the patients’ capacities to sustain meaningful and purposeful interactions due to a relevant dysfunction in the psychological

activity (for instance, dysfunctions in cognitive processes, emotional dysregulation, abnormalities in the sense of agency or in the sense of ownership over the patients' subjective states, and so on).

C. The diagnostic evaluation involves a qualitative judgment about values. A harmful dysfunction analysis has been proposed as a rational link between scientific facts and social values, and as a foundation for psychopathology (Wakefield 2017). Psychopathological constructs rely on facts as much as they rely on values, but this is a common feature of all medical constructs. The normative judgments about harmfulness are an integral part of the diagnostic process leading to therapeutics both in the field of mental health and in the field of general medicine (Jefferson 2022). To be valid, the assessment should consider if there is an intrinsic relationship between the cluster of mental symptoms/behavioral signs, and the states of suffering and/or harmful dysfunction, and not merely an effect of extrinsic factors like stigma and discrimination. The social, political, cultural, economic, and interpersonal sources of distress should be assessed before establishing the diagnosis of a psychopathologic pattern. These may be present in many cases, but the validity of a psychopathological diagnosis can be established only if the harmfulness is intrinsically related to the psychological dysfunction (Wakefield 2017; Jefferson 2022).

5 Neuropsychiatric Constructs

In the clinical setting, how should we decide if a patient has a neuropsychiatric problem? Neuropsychiatric constructs are not fixed objects with a perfectly stable ontology, but instead, pragmatic diagnostic concepts allowing for a rational connection between the level of clinical care and the level of scientific research. Neuropsychiatric constructs must fulfill both the criteria previously exposed for neurological constructs and for psychiatric constructs, but also, a significant relationship between the psychopathological pattern and a specific neurologic condition should be established. How can we know if it is not merely a casual or a trivial coexistence of two independent phenomena? The scientific answer to that question involves the longitudinal study of the individual case, a clinical epidemiology approach, and a clinical neuroscience approach, given that the clinician needs a scientific theory of the relationship between the psychopathological events and the neuropathological patterns.

5.1 Brain-Behavior Relationships and the Construction of a Neuropsychiatric Diagnosis

Complex behavior is not the result of a set of inherited sensory-motor reflexes. If the behavioral response is fixed through direct connections between sensory and motor neurons, without reaching the level of intermediary processing in the brain, we observe the stereotyped pattern of reaction that characterizes nervous reflexes (for example, the stretch reflex). When it comes to complex forms of behavior, sensory inputs are subject to intermediary processing before reaching motor neurons

(Mesulam 2000b). This comprehends the association neocortical structures, the paralimbic cortices, the allocortical structures, and subcortical nuclei traditionally encompassed as parts of the cortico-limbic and cortico-striatal circuits, as well as the cortico-cerebellar circuits. This is always achieved through the integrated activity of functional networks. As the psychological level of description, we observe adaptive, flexible patterns of behavior through which the organism interacts in different ways with an everchanging environment, according to the individual's learning history (Mesulam 2000a; Peña-Casanova et al. 2024). The interactions between the organism and the environment induce physiological and structural changes by means of neuronal plasticity mechanisms (synapse formation, long term potentiation, long term depression, neurogenesis) (Ramirez 2018; Asok et al. 2019). These structural modifications are necessary to stabilize the behavioral changes obtained through learning.

Since the late XIX century, clinico-pathological relationships were established between the level of behavior and the existence of structural brain lesions. Cognitive and affective problems were observed after focal lesions: amnesia as related to damage in the hippocampus, disinhibited behavior after lesions of the ventral parts of the frontal lobe, expressive aphasia in patients with lesions of the third frontal gyrus in the left hemisphere, or receptive aphasia in patients with lesions of the left parieto-temporal junction (Penfield and Milner 1958; Mesulam 2000a). Soon, other focal neuropsychological syndromes were described: alexia, agraphia, acalculia, Balint's syndrome, visual agnosia, amusia, ideomotor apraxia, and many others. These syndromes may be subject to a critical review of the behavioral phenotype and its relationship to brain networks, and controversies may appear regarding the concepts of functional specialization and functional integration in the context of psychological activity (Westlin et al. 2023). However, they are helpful resources for the clinical understanding of brain-behavior relationships.

How do we establish valid and reliable relationships between psychopathological patterns and neurological pathologies? If both are present in a given case, how do we know that a significant relationship exists, instead of a random coexistence? The recognition of focal neuropsychological syndromes, like alexia, agraphia, acalculia, and so on, provides a reliable clue to proceed with neurological investigations in a specific case. Consider the case of a 29-year-old male patient with right hand dominance, who has an acute cognitive disturbance with psychotic symptoms, followed by a disturbance in speech production, with reduced verbal output, grammar deficiencies, and preserved language comprehension. The brain-behavior relationships established by clinical neuroscience dictate that the clinician must solicit a brain imaging study to rule out a structural lesion affecting Broca's and surrounding areas. Figure 2 presents the brain image. If focal neuropsychological syndromes are present, the knowledge of brain-behavior relationships imposes a deontological imperative to assess through clinical and technological resources the hypothesis of an underlying neuropathological abnormality. Focal neuropsychological syndromes are the less controversial problems within the territory of neuropsychiatry and cognitive neurology. However, in the case presented in Fig. 2 the emergence of psychotic symptoms is not predicted by the focal lesion. This reminds us that most psychopathological syndromes, like psychosis, delirium, mania or catatonia, are not well explained in terms of focal lesions and isolated defects of brain functions. This leads to alternative frameworks consider-

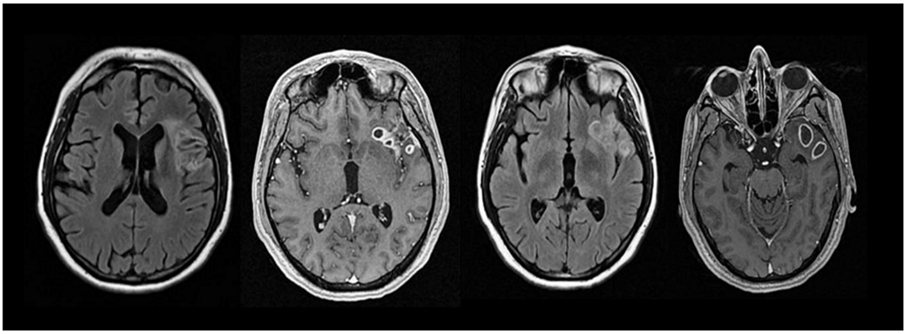


Fig. 2 A neuropsychiatric clinical case that aligns with well-known brain-behavior relationships. A 29-year-old right-handed male patient presents with a newly onset speech disorder, progressing to a fluctuating state of inattention, disorientation, visual hallucinations, and psychomotor agitation. The patient has a history of a recent dental infection with the extraction of a tooth. Magnetic Resonance Imaging reveals a localized lesion in the left hemisphere, affecting the opercular region of the frontal lobe, the insular cortex, and the anterolateral temporal lobe, which is consistent with an aphasic syndrome. The EEG shows a pattern of generalized slowing, which is consistent with the progression to a state of delirium. A bacterial abscess was diagnosed, and antibiotic treatment was successful. During the recovery, a careful neuropsychological examination reveals a preserved capacity to understand language, but a significantly reduced verbal production, with multiple grammatical errors. A complete recovery was achieved in the long term

ing the role of connectomics and distributed neural ensembles giving rise to mental events, rather than explanations relying on localized neural populations (Westlin et al. 2023). Also, instead of explaining psychiatric symptoms and syndromes as direct manifestations of pathobiological factors, we must consider the effect of abnormal brain signals on the dynamics of distributed cognitive and affective systems. Cognitive models are essential for understanding the emergence of psychopathological patterns in patients with brain disorders (Halligan and David 2001). Thus, a look at the difficult cases is much needed.

5.2 The Epistemological Feedback Between Clinical Practice and Neuroscientific Research: Beyond Focal Neuropsychological Syndromes

During the last century, focal neuropsychologic syndromes, tied to brain lesions, have provided a robust framework to understand brain-behavior relationships. However, there are neuropsychiatric patterns of severe cognitive dysfunction (e.g., delirium or rapidly progressive dementia) which are not as the result of a focal lesion, but instead they are conceived as manifestations of a diffuse, generalized encephalic dysfunction (encephalopathy) (Trzepacz et al. 1985; Oldham and Holloway 2020). Also, many psychiatric signs, symptoms and syndromes are not well explained in terms of focal or generalized demonstrable deficits. Psychosis, catatonia, mania, depression, anxiety, obsession-compulsion, dissociative phenomena, aggressive behavior, and many other psychiatric phenomena may be seen in neurological patients but also in patients without a demonstrable neurological condition. This fact presents a difficulty at the theoretical and practical levels. If a clinical pattern usually identified with primary psychiatric disorders (for instance, schizophrenia-like psychosis) is observed in a patient with a neurological condition, how can it be established that there is a significant relationship between the psychiatric and the neurological facts? As mentioned above, there are three main approaches to answer that question: a clinical epidemiology approach, a clinical neuroscience approach, and a longitudinal study of the individual case. When the three approaches converge, there is a gain in clinical certainty.

5.2.1 The Clinical Epidemiology Approach

It is well-known that individual cases only rarely provide sufficient information on matters of causality. The clinical epidemiology approach is based in the group analysis of similar cases and controls subjects, allowing for inferential statistics, improving our capacity to establish causal inferences beyond biases, confounding variables, and chance. For instance, the case of Augusta D. -which lead to the clinico-pathological description of Alzheimer's disease- shows that she suffered from delusions one year before the cognitive deficit was observed (Ramirez-Bermudez 2012). There is a scientific consensus regarding the causal relationship between Alzheimer's neuropathology and the progressive cognitive decline in patients with Alzheimer's type-dementia. Both the neuropathological abnormalities and the cognitive dysfunction are required for a definite diagnosis of Alzheimer's disease. But delusions are present only in some cases. The question would be: was Alzheimer's pathology the cause of Augusta D delusions? By looking at her individual case, it is impossible to reach a conclusion. However, clinical epidemiology studies have shown that delusions are significantly more common in patients with Alzheimer's disease (22%) as compared to age matched individuals without dementia (2%) (Leroi et al. 2003). This statistical relationship has been confirmed by multiple studies (Cummings et al. 2020). By using the epidemiological definition of cause (Elwood 2009), it is reasonable to support the hypothesis of Alzheimer's pathology as a causal factor of delusions in the elder, even if the process of symptom formation is not perfectly established and involves multicausality, and even if Alzheimer's pathology is not a necessary and sufficient cause, because: (a) not all patients with Alzheimer's disease present delusions, and (b) there are other causes of delusions in the elder. However, a mechanistic model to understand these relationships is useful as a conceptual tool for hypothesis formation and testing.

5.2.2 The Clinical Neuroscience Approach

According to the classic Bradford Hill criteria (Fletcher et al. 2014), a strong theory constructed by means of empirical data provides a logic and mechanistic framework for biological plausibility, which is a relevant criterion in the assessment of causation. In the context of neuropsychiatric constructs, clinical neuroscience provides a plausibility framework and sometimes a strong causal model. The gradual construction of an integrated theory about the relationship between the nervous system and the psychological activity requires neuroscientific studies done in healthy subjects, in patients with brain lesions, and in subjects receiving a diagnosis of primary psychiatric disorders. This integrated theory is necessary to approach the difficult cases that lie in the borderland between neurology and psychiatry. Consider Fig. 3, which

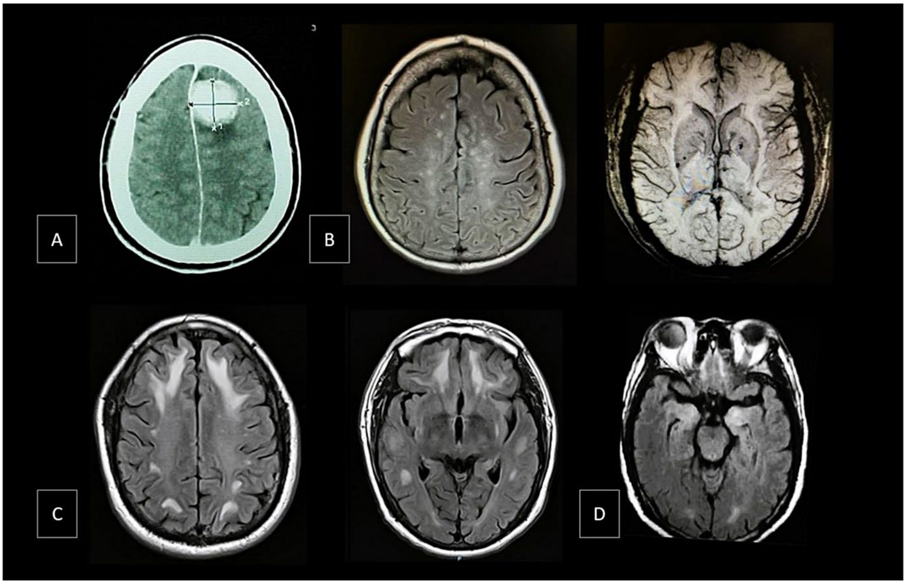


Fig. 3 Examples of cases in the borderland between neurology and psychiatry. In case A, a 46-year-old woman develops a 6-month progressive state of psychotic depression, with a marked slowing of psychomotor activity, and a reduction in goal directed behavior and spontaneous speech. The CT scan reveals a brain tumor with surrounding edema affecting the left dorsomedial cortex. Case B refers to a 60-year-old woman with a post-COVID neuropsychiatric condition characterized by cognitive impairment and depression. Her MRI in FLAIR sequence shows multiple hyperintense images in the semioval center, bilaterally, corresponding to punctate gliosis. In the SWI sequence, hypointense images are observed in the right putamen and the left globus pallidus, corresponding to the sequelae of microhemorrhages. Case C refers to a 49-year-old man with a history of chronic exposure to solvents who develops a state of disorientation, memory disturbance, confabulation, complex visual hallucinations, and euphoric affect. Brain MRI revealed bilateral white matter lesions predominantly in the frontal lobes. Case D refers to a young woman with anti-NMDA receptor encephalitis. The MRI shows a hyperintense signal in the medial temporal lobes, bilaterally

shows examples of cases that appear in the borderland between neurology and psychiatry. In **case A**, a 46-year-old woman without a psychiatric history or relevant psychosocial risk factors develops a 6-month progressive state of severe depression, with a marked slowing of psychomotor activity, and a reduction in goal directed behavior and spontaneous speech. She describes a profound state of sadness, hopelessness, and delusional guilt. In the mental state examination, there is a significant reduction in planning, sequencing, working memory and retrograde memory. The CT scan reveals a brain tumor with surrounding edema affecting the left dorsomedial cortex. The brain topography explains why she had features of a dysexecutive syndrome, but it is unclear why she had a severe state of psychotic depression. New perspectives of neuroscience such as connectomics suggest that mental function is the product of the activity of a set of dynamically interconnected brain regions, which provides a biological framework to tie the dysfunction in brain networks to psychiatric symptomatology (Deco and Kringelbach 2014; Cao et al. 2015). Functional neuroimaging studies suggest that a dysfunction of brain networks related to prefrontal cortex is relevant to psychosis, depression, and more specifically to the emergence of psychosis in patients with severe depression (Busatto 2013). This provides a clue for the interpretation of **case A**. Thus, research of primary psychiatric disorders may provide a mechanistic framework of psychobiological plausibility that is useful to interpret specific neuropsychiatric cases. In return, neuropsychiatric cases may be conceived as milestones for the construction of an epistemological feedback between clinical practice and psychiatric neuroscience. This feedback contributes to stronger theories beyond focal neuropsychological syndromes. To better grasp the interaction between different levels of evidence in clinical neuroscience, it is vital to differentiate between structural imaging of an individual case, which identifies a distinct, localized abnormality in the patient's brain that contributes to understanding the symptoms, and findings from functional and structural brain imaging derived from larger datasets in psychiatric neuroscience. These studies involve analyzing data from patient groups compared to control groups to infer dysfunctional brain networks. This approach provides an evidence-based explanatory framework for symptoms observed in a specific case.

Case B refers to a 60-year-old woman with a post-COVID neuropsychiatric condition. Two close family members died from complications related to COVID-19. During the acute phase of her infectious disease, she experienced significant hypoxia, with oxygen saturation levels of 60%, and acute mental status changes consistent with delirium and encephalopathy. In the post-acute phase, she developed a severe state of depression, which did not respond to psychological treatments. Her neurocognitive evaluation revealed severe impairments in processing speed, working memory and in the ability to retrieve information in tasks involving semantic memory, episodic memory, and visual memory. Figure 3 shows multiple hyperintense images in the semioval center, bilaterally, corresponding to punctate gliosis, and sequelae of microhemorrhages in the right putamen and the left globus pallidus. The case is consistent with clinical neuroscience research showing the neuropsychiatric effects of the COVID-19 infection, especially when there is encephalopathy during the acute phase and pre-existing medical and psychiatric comorbidities. Among the common neuropsychiatric sequelae are depression, anxiety, and cognitive impairment. In the acute phase, cerebrovascular complications are common due to endothelial damage leading to the formation of microthrombi, and inflammatory processes mediated by interleukins and microglial activation (Garg 2020; Rogers et al. 2020; Boldrini et al. 2021; Lopez-Leon et al. 2021; Taquet et al. 2021). The psychopathological pattern is constituted by depression and cognitive impairment, the two of which may be related through a bidirectional or circular causation process, as depressive behavior may increase the cognitive dysfunction, whereas cognitive dysfunction contributes to the impairment in social functionality that can worsen the depressive state. Overall, the explanation of the psychopathological pattern involves both psychosocial factors, as the traumatic experience of the acute infection, and the loss of two family members in the same timeframe, but psychobiological factors as well fronto-parietal white matter damage, and neuroinflammation leading to changes in putamen, psychomotor slowing, anhedonia and anergia (Braga et al. 2023).

Case C refers to a 49-year-old man with a history of chronic exposure to solvents. His illness began six months before hospitalization, with cognitive and emotional changes, and aggression towards family members. Upon admission, he was severely disoriented. He claimed to be "ninety-eight years old" and "hospitalized for a hundred years." He denied being in a hospital in Mexico City and insisted he was in his "beloved town of Nicolás Romero." He reported experiencing complex visual hallucinations of unfamiliar people. He displayed a euphoric affect, making jokes and claiming to feel "really great." Brain MRI revealed bilateral lesions in the white matter of the frontal lobe. In this case, the topography helps to understand the disturbance in memory retrieval and the phenomenon of confabulation. But the state of euphoria and the content of the mental states (he believes he is in his "beloved" hometown) is consistent with the hypothesis that the content of confabulation is emotionally biased. Previous experimental work has shown that "motivational factors, along with defective reality and temporality monitoring, contribute to confabulation." (Fotopoulou et al. 2007) This may lead to "wishful reality distortions" in confabulation, and more specifically in confabulatory false beliefs about place (Fotopoulou et al. 2004; Turnbull et al. 2004) at this point, it is worth to consider the issue raised by German Berrios: even in the patients with well-defined neurological conditions, the psychosocial and biographical variables have an influence over certain symptoms that are not fully explained by mechanical brain-behavior relationships. "Neurological patients have reasons for their symptoms, that is, neurological diseases happen to real people and hence have semantic contexts. This adds an entire new layer of meaning, hermeneutics and therapeutic response" (Berrios 2007). Furthermore, the emergence of psychiatric symptoms like confabulation, delusions or hallucinations calls for multidisciplinary explanations. According to the cognitive neuropsychiatry approach, which offers a conceptual framework for explaining psychopathology, this is achieved by linking mechanistic knowledge of brain networks with a functional approach to cognitive systems. This approach provides a bridge between the neurobiological level and the psychiatric symptom level through the resource of cognitive models.

5.2.3 The Longitudinal Study of the Case

While some neuropsychiatric cases are well explained by the current science of brain-behavior relationships, other cases located at the borderlands between neurology and psychiatry inconsistent with the reductionist view that phenomenological and psychological manifestations can be replaced by mere neurophysiological patterns. Consider **case D** in Fig. 3. This refers to a woman who suffered early abuse in her childhood, and developed nonsuicidal self-directed violence (cutting). Years later, she suffers from a severe episode of psychosis and catatonia, progressing to delirium with seizures. A diagnosis of definite anti-NMDA receptor encephalitis is made after a positive determination of the antibodies in the cerebrospinal fluid, an MRI showing a hyperintense medial temporal lobes, and an EEG with generalized slowing. After immunotherapy, a significant response is observed. But in the long term follow up, the patient has relapses of psychosis and emotional dysregulation with suicidal thoughts and behaviors, exacerbated by interpersonal stress and reminiscences of the sexual abuse. The possibility of relapses of the autoimmune pathology are ruled out through diagnostic testing. The hypothesis is that her atypical outcome is related to the background of traumatic experiences, and the long-term effects of encephalitis.

In a broader sense, we postulate that psychosocial and neurobiological causal factors interact in many cases, even if there is clinical evidence of a neurological disease related to psychopathology. This is consistent with Engel's biopsychosocial model, in general terms. Although the classic formulation by Engel lacked the details that are required to account for neuropsychiatric constructs, the more recent formulation of the model by Bolton & Gillet is quite helpful, mainly because they conceive three thesis which we endorse: Firstly, psychological activity is embodied agency. Secondly, the conditions for embodied agency are biopsychosocial. Thirdly, the lack of biopsychosocial conditions for the development of agency jeopardies the health of the organism, including its mental health (Bolton and Gillett 2019). This calls for a transdisciplinary approach, including not only the study of the neuropathological and psychopathological dimensions, but also the study of contextual variables, idiographic trajectories, and a rigorous approach to the content of mental states and the personal meaning of the symptoms. A subpersonal, mechanistic account of pathophysiological processes is considered to support the biological plausibility of a scientific theory of causation in neuropsychiatry. However, psychological mechanisms need to be taken into account in the case of mental health disorders, even if these mechanisms are not framed through a bio-reductionist discourse (Borsboom et al. 2019). A theoretical bridge has been conceived to connect mechanistic and subpersonal causal explanations with psychological explanations, through the understanding of coherent relations from parts to whole and from whole to parts. Coherence is grounded in interdependent processes conditioned by learning, neurobiological mediating variables, and environmental contexts, which impose constraints on the nervous system activity (Juarrero 2023).

6 Closing Remarks

This article aimed at explaining the logic of neuropsychiatric constructs. Even if superficially, we proposed a definition of the pathological, and we approached neuropathological constructs as well as psychopathological constructs, leading us to conceptualize structural lesions, physiologic abnormalities, behavior, subjective experience, neuropsychological functions, as well as psychopathological patterns. The study and the clinical care of neuropsychiatric cases require a proper recognition of both psychopathological and neuropathological patterns. Moreover, it entails an explanation about the relationship between these patterns based on the best scientific evidence about pathophysiological mechanisms and epidemiological relationships. We discussed the logic of brain-behavior relationships and the need for closer academic feedback between neuroscientific research and clinical practice. Some neuropsychiatric cases are well explained by the current science of brain-behavior relationships. But hybrid cases located at the boundaries of this science require a transdisciplinary approach, leading to a study of the complexity of interactions, the temporal trajectories, the personal meaning of symptoms, and the study of sociocultural contexts. While the primary goal of this article is to clarify and explain neuropsychiatric constructs, we also advocate for an integrative conceptualization of neuropsychiatric disorders, positing that psychosocial factors must be understood through a cognitive, affective and social neuroscience perspective in order to understand their role as causal factors for psychopathology. This approach necessitates a translational science capable of addressing how social stress and histories of psychological interaction alter the organism's state, particularly the neurobiological systems that underlie psychological states and activity. This perspective can be effectively explained using an allostatic model (McEwen 1998; Schulkin and Sterling 2019; Shaffer et al. 2022).

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Data availability As this is a theoretical work, no empirical data were generated during the study.

Declarations

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